ABOUT ME

ASHIRVAD

My Journey, Personality, Education, Skills,
Interests, Goals and Aspirations

A personal profile and self-reflection

Prepared from information shared in conversations

PAGE 1 – INTRODUCTION: WHO AM I?

My name is Ashirvad. I am a Computer Science and Engineering student who is interested in technology, programming, artificial intelligence, software development, and continuous learning.

I consider myself someone who is still exploring different areas of technology while gradually building the knowledge and skills required to establish a successful career in the IT industry. My journey so far has been a combination of academic learning, practical experimentation, problem-solving, and discovering what I genuinely enjoy doing.

One of the most important aspects of my personality is my curiosity. I don't always feel satisfied with simply knowing the definition of something. I generally want to understand how it works, why it works, and how I can implement it myself.

For example, when learning web development, I don't just want to know what React is. I want to understand how React works, how components communicate, how applications are structured, how data flows, and how a project eventually becomes a working website.

Similarly, while learning artificial intelligence, I am interested in understanding the mathematical and theoretical foundations behind algorithms rather than simply using existing libraries.

I believe that technology is not just about learning programming languages. It is about developing the ability to identify problems, understand their underlying structure, and create practical solutions.

I am currently pursuing my education in Computer Science and Engineering. My academic interests include programming, data structures, algorithms, artificial intelligence, machine learning, database management systems, computer architecture, and software development.

Apart from academics, I am interested in exploring modern technologies, building projects, improving my technical abilities, and understanding the changing requirements of the software industry.

I don't consider my learning journey complete. In fact, I see myself as someone who is still developing the foundation for a much bigger professional journey.

My objective is to become technically strong, independent in problem-solving, capable of building real-world applications, and confident enough to compete in the software development industry.

PAGE 2 – MY EDUCATIONAL BACKGROUND AND ACADEMIC INTERESTS

I am pursuing a Bachelor of Technology (B.Tech) in Computer Science and Engineering, with an academic connection to Uttarakhand Technical University. Computer Science is a broad field, and my academic journey has exposed me to several important areas of computing.

Some of the subjects and topics I have explored include:

- Data Structures and Algorithms; Database Management Systems; Object-Oriented Programming; Computer Organization and Architecture; Artificial Intelligence; Machine Learning; Natural Language Processing; Probability and Statistics; Operating Systems; Computer Networks; and Web Development.

I have particularly explored Artificial Intelligence and Machine Learning in considerable detail. My AI/ML learning interests include supervised learning, unsupervised learning, classification, regression, clustering, dimensionality reduction, probabilistic reasoning, neural networks, reinforcement learning, and evolutionary algorithms.

I have also studied concepts such as:

- Decision Trees and CART; Random Forests and Ensemble Learning; Bagging and Boosting; K-Nearest Neighbors; K-Means Clustering; Gaussian Mixture Models; Principal Component Analysis; Linear Discriminant Analysis; Independent Component Analysis; Genetic Algorithms; Fuzzy Logic; Bayesian Networks; Hidden Markov Models; and Conditional Random Fields.

My interest in mathematics has also influenced my attraction towards AI and machine learning. I have previously described myself as quite good at mathematics, which makes mathematical concepts behind machine learning particularly interesting to me.

I prefer understanding the logic behind algorithms instead of memorizing their definitions. For example, when studying the Euclidean Algorithm, I wanted to understand the fundamental mathematical idea behind finding the greatest common divisor rather than simply remembering the algorithm.

Likewise, when studying probability and Bayesian reasoning, I wanted to understand how probabilities influence decision-making.

My academic approach is generally based on understanding concepts step by step, connecting theory with practical examples, and then applying that knowledge to programming or problem-solving.

I also recognize that academic knowledge and industry-ready skills are not exactly the same. Therefore, I am interested in combining my university education with practical projects, coding practice, internships, and independent learning.

PAGE 3 – MY PROGRAMMING JOURNEY AND TECHNICAL SKILLS

Programming is one of the most important parts of my technical journey. I have explored multiple programming languages and technologies, gradually moving from basic programming concepts towards more advanced development topics.

1. C and C++

I have experience with C and C++, and I have used C++ extensively for Data Structures and Algorithms practice. Some of the topics I have explored include:

- Arrays and strings; searching and sorting; linked lists; stacks and queues; recursion; sliding window techniques; two-pointer techniques; mathematical algorithms; time and space complexity; and LeetCode problem-solving.

I particularly enjoy working on problems involving linked lists and arrays. I have worked on problems such as Add Two Numbers, Swap Nodes in Pairs, Odd-Even Linked List, Delete Duplicates, Maximum Product of Two Elements, and Minimum Recolors to Get K Consecutive Black Blocks.

My preferred learning approach is to attempt the problem myself before looking at a complete solution. When I encounter a mistake, I generally want to understand exactly where my logic fails. I also prefer explanations that preserve my original approach instead of replacing my entire solution with a completely different implementation.

2. Python

Python is another language I have explored, particularly because of its applications in artificial intelligence, machine learning, automation, and backend development. My learning has included variables and data types, functions and loops, object-oriented programming, classes and inheritance, polymorphism, multiple inheritance, Method Resolution Order, magic methods, threading and multiprocessing, asynchronous programming, and Python libraries for machine learning.

I have also explored practical Python case studies, including a campus management system and an attendance processing system.

3. HTML and CSS

My web development journey began with HTML and CSS. I learned how websites are structured, how elements are styled, and how different sections of a webpage are organized. I have worked with HTML document structure, forms, CSS selectors, Flexbox, page styling, external CSS linking, and basic responsive design concepts.

4. JavaScript

After HTML and CSS, I started learning JavaScript to make websites interactive. My learning included DOM manipulation, events, functions, promises, callbacks, asynchronous programming, and async/await. I also created small projects such as a To-Do application, a number guessing game, and DOM-based interactive exercises.

5. React

React is one of the major technologies I am currently exploring. I have worked with React components, JSX, ReactDOM, Parcel, project configuration, and the development environment. I have also encountered practical issues involving imports, exports, package.json configuration, JSX compilation, React list keys, and Git integration.

PAGE 4 – MY INTEREST IN WEB DEVELOPMENT AND SOFTWARE ENGINEERING

Web development is one of my major technical interests. I want to develop the ability to create complete applications, from the frontend interface to the backend logic and database.

My journey started with basic HTML and CSS and gradually moved towards JavaScript and React. I have also explored different career paths within web development. For example, I have considered learning the MERN stack and Java Full Stack Development. I am interested in understanding which technologies offer stronger long-term opportunities and which learning paths can help me become job-ready.

My React learning journey

I started exploring React through a project named Namaste React. The project helped me understand how a modern JavaScript development environment works. I worked with React, ReactDOM, JSX, Parcel, npm, package.json, Git, and GitHub.

I encountered several issues while configuring the project, but gradually resolved them. This experience introduced me to real development workflows rather than only theoretical examples.

My interest in full-stack development

I am interested in understanding both frontend and backend development.

- Frontend: HTML, CSS, JavaScript, React, and modern UI development.
- Backend: Java or Python, APIs, server-side programming, authentication, and database integration.
- Development tools: VS Code, Git, GitHub, npm, and Parcel.

I also want to understand how AI can be integrated into web applications. The emergence of AI-assisted development has made me curious about how developers can use AI tools to build applications more efficiently.

However, I want to develop the underlying programming knowledge rather than depend entirely on AI-generated code.

My long-term objective is to become capable of developing complete, functional, and practical software applications independently.

PAGE 5 – MY INTEREST IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Artificial Intelligence and Machine Learning are among the most interesting areas of technology for me. I am particularly attracted to the combination of mathematics, algorithms, programming, and intelligent decision-making.

I have explored both traditional AI concepts and modern machine learning approaches.

Machine Learning

I have studied supervised learning, unsupervised learning, regression, classification, clustering, and dimensionality reduction. I am interested in understanding how models learn from data and how their performance can be evaluated.

Natural Language Processing

NLP is another area that interests me. Some topics I have explored include Part-of-Speech Tagging, Penn Treebank, Hidden Markov Models, Maximum Entropy Models, Conditional Random Fields, context-free grammars, parse trees, semantic analysis, lexical ambiguity, homonymy, and polysemy.

I find it interesting that computers can process human language and extract meaningful information from text.

Probabilistic Reasoning

I have explored probability theory, conditional probability, Bayes' theorem, and Bayesian networks. I am interested in understanding how AI systems make decisions when information is incomplete or uncertain.

Neural Networks and Modern AI

I have also explored introductory neural network concepts, reinforcement learning, reward models, and transformer-based models. My practical interest in AI has extended to building applications rather than studying algorithms alone.

I have worked on the idea of an AI and NLP Concept Dashboard using a Flask backend and a React-based frontend. The project involved exploring reward scoring, comparing two responses, using transformer models, and integrating machine learning functionality into a web interface.

This project represents an important direction in my learning: combining artificial intelligence with software development.

I want to become someone who understands both how AI models work and how they can be integrated into useful applications.

PAGE 6 – MY PROJECTS AND PRACTICAL EXPERIENCE

Practical projects are an important part of my learning process. I believe that implementing concepts is one of the best ways to understand them. Over time, I have explored several projects across programming, web development, and artificial intelligence.

1. Namaste React

This is one of my React learning projects. I worked on setting up a React environment using Parcel, React, and ReactDOM. I explored components, JSX, project configuration, package management, and GitHub integration. The project also introduced me to practical issues involving build tools and dependency management.

2. Task Flow – To-Do Application

I explored a task management application called Task Flow. The application included adding tasks, displaying tasks, deleting tasks, editing tasks, updating existing tasks, and marking tasks as completed. The project helped me understand DOM manipulation and how user interactions can update the interface.

3. Simon Says Game

I also explored developing a Simon Says memory game. This project is useful for understanding JavaScript events, sequences, user input, game logic, and interactive interfaces.

4. AI and NLP Concept Dashboard

One of my more ambitious project ideas involves combining AI/ML functionality with a web application. The project uses a Flask backend and a React-oriented frontend. Its functionality includes comparing responses, generating reward scores, and presenting AI-related information through a dashboard.

I have explored Python, Flask, CORS, PyTorch, NumPy, Matplotlib, scikit-learn, and Hugging Face Transformers in connection with this project.

5. Shivalik Campus Management System

I have also explored an object-oriented Python case study involving a campus management system. The proposed system includes Student, Faculty, Staff, Department, Course, and Hostel entities. The project provides an opportunity to understand inheritance, polymorphism, multiple inheritance, MRO, and design patterns.

6. Campus Attendance Processing System

Another Python case study involved processing attendance information using threading, multiprocessing, asyncio, and executors. This helped me explore concurrency and parallel execution.

7. DBMS Practical Work

I have also worked with database-related concepts involving student and course tables, relational algebra, SQL queries, and triggers.

These projects and exercises reflect my interest in learning through implementation. I want to gradually move from small educational projects towards larger applications that solve genuine problems.

PAGE 7 – MY PROBLEM-SOLVING APPROACH AND LEARNING PERSONALITY

One of the most important things I have discovered about myself is that understanding a solution and independently discovering a solution are two different abilities.

While practicing Data Structures and Algorithms, I have sometimes understood an optimized solution after seeing its explanation but struggled to independently identify the same approach. This has made me realize that problem-solving requires a separate kind of training.

I am working towards developing stronger logical thinking, pattern recognition, and algorithmic intuition.

My preferred learning process generally involves:

- 1. Understanding the problem statement; 2. identifying the important conditions; 3. thinking about a possible approach; 4. writing my own code; 5. identifying errors; 6. understanding why those errors occur; 7. learning the optimized approach; and 8. practicing similar problems.

I particularly appreciate explanations that break down code into smaller parts. For example, when learning linked lists, I prefer understanding what a pointer represents, where it points, and how changing its next pointer affects the entire list.

I also like understanding individual C++ expressions and functions rather than treating them as mysterious syntax. I frequently explore concepts such as auto, lower_bound, find, stoll, ternary operators, stack top, and pointer manipulation.

My learning personality can be described through several characteristics:

Curiosity

I frequently explore new concepts and technologies.

Practical thinking

I prefer seeing how concepts work in actual applications.

Persistence

I continue working through technical errors and debugging problems.

Detail orientation

I often ask for explanations of individual lines and concepts.

Exploration

I am interested in multiple areas of computer science rather than limiting myself to one subject.

I also recognize that I need to keep improving my consistency, independent problem-solving ability, and depth of technical understanding. I see these as areas for development rather than limitations.

PAGE 8 – MY INTERESTS OUTSIDE PROGRAMMING

Although technology occupies an important part of my learning journey, my interests are not limited to programming. I am also interested in exploring different subjects, creative possibilities, and practical skills.

1. Technology and Emerging Trends

I like understanding how new technologies work and how they affect the software industry. I have explored questions about AI-assisted content creation, AI-generated videos, AI-powered development tools, and the changing nature of web development.

I am particularly curious about how a written script can be transformed into a complete video using AI tools. This reflects my interest in the creative applications of technology.

2. Creative Storytelling

I have explored historical and fictional storytelling, particularly involving the world of Game of Thrones and the Robert's Rebellion storyline.

I have shown interest in creating scene-based scripts with detailed character appearances, personalities, physical descriptions, and cinematic presentation. I like the idea of transforming written stories into visual content using AI tools.

This combines creativity, storytelling, technology, and visual imagination.

3. Personal Development

I am also interested in improving myself outside academics. My questions have included fitness, exercise routines, skincare, hair care, and general lifestyle improvement.

These interests reflect an awareness that professional development is only one part of personal growth.

4. Learning New Skills

I have also explored interests such as learning guitar and understanding new software tools. I enjoy discovering skills that can expand my abilities beyond my primary academic field.

Overall, I would describe my interests as a combination of technology, creativity, self-improvement, and practical experimentation.

PAGE 9 – MY CAREER AMBITIONS AND FUTURE GOALS

My major professional objective is to establish myself in the technology industry. I want to develop strong technical skills, build practical projects, gain internship experience, and eventually secure a good software development opportunity.

I have explored several career directions, including Software Development, Full-Stack Web Development, Java Development, React Development, Artificial Intelligence, Machine Learning, and Natural Language Processing.

I am still exploring the different possibilities and trying to understand which direction aligns best with my interests and long-term goals.

My short-term goals

- 1. Strengthening my C++ and DSA knowledge; 2. improving JavaScript fundamentals; 3. learning React properly; 4. understanding backend development; 5. building complete web applications; 6. improving Git and GitHub skills; 7. developing practical AI/ML knowledge; 8. preparing for internships and technical interviews; and 9. improving my understanding of databases and software engineering.

My medium-term goals

- Become comfortable developing full-stack applications; build projects that demonstrate real technical ability; develop a strong GitHub portfolio; gain practical internship experience; improve coding interview performance; understand software architecture and application development; and explore the integration of AI into software products.

My long-term goals

My long-term ambition is to become a skilled software engineer who can independently design, develop, debug, and maintain real-world applications.

I want to understand technology deeply enough to make good engineering decisions rather than simply following tutorials. I also want to keep learning as technology evolves.

The software industry changes continuously, and I believe adaptability will remain one of the most important qualities for a successful developer.

I want to build a career based on strong fundamentals, practical knowledge, and the ability to solve meaningful problems.

PAGE 10 – MY STRENGTHS, AREAS OF IMPROVEMENT AND PERSONAL VISION

Looking at my learning journey, I can identify several strengths that I want to continue developing.

My strengths

1. Curiosity and willingness to learn

I am interested in exploring new technologies and understanding how they work.

2. Mathematical interest

My interest in mathematics makes me particularly attracted to algorithms, AI, probability, and machine learning.

3. Practical learning orientation

I enjoy implementing concepts through projects and coding exercises.

4. Technical exploration

I have explored multiple areas, including frontend development, backend development, AI/ML, databases, and programming.

5. Debugging and experimentation

I have worked through real development errors involving React, Parcel, Git, Python, Flask, and project configuration.

6. Desire for conceptual clarity

I prefer understanding why something works instead of blindly memorizing an answer.

Areas where I want to improve

- Becoming more independent at solving DSA problems; maintaining consistency in daily learning; developing deeper knowledge of one primary technology stack; completing more projects instead of continuously starting new ones; improving software engineering practices; building confidence in technical interviews; improving communication and technical explanation skills; and learning how to convert ideas into finished products.

I believe identifying weaknesses is important because it provides direction for improvement.

My personal vision

When I look at my overall journey, I see someone who is exploring, learning, experimenting, and gradually developing a professional identity.

I have interests in multiple areas of computer science, but I understand that real expertise requires focus and consistent practice.

I don't want to become someone who only knows the names of technologies. I want to become someone who can actually use them to create something valuable.

I want to understand the fundamentals of programming, become strong at problem-solving, build useful applications, and understand the mathematics and logic behind artificial intelligence.

I also want to retain my curiosity and willingness to explore new ideas. I understand that the journey will involve mistakes, confusion, debugging, failed attempts, and continuous improvement. But I see these experiences as necessary parts of becoming a capable engineer.

My ultimate objective is to become a technically strong, creative, adaptable, and independent software professional who can turn ideas into working solutions.

CONCLUSION

I am a Computer Science student with interests spanning software development, artificial intelligence, machine learning, programming, problem-solving, and creative technology.

My journey reflects a desire to understand technology from its fundamentals to its practical applications. I am still learning, still experimenting, and still discovering the direction in which I want to specialize.

But one thing remains consistent: I want to keep improving my knowledge, develop meaningful skills, and build things that demonstrate what I am capable of achieving.

I am not defined only by what I have learned so far, but also by what I am willing to learn and accomplish in the future.